COL724/7524 Advanced Computer Networks

Assignment 1

Part B: Direct Peering and Path Comparison

Objective

The objective of Part B is to study the effect of adding a direct peering link between AS1 and AS4 to the inter-AS topology.

The baseline topology from Part A contains the chain

$$AS1 \to AS2 \to AS3 \to AS4.$$

In Part B, a new direct link is added between AS1 and AS4:

$$AS1 \to AS4.$$

The new link is configured with higher bandwidth and lower propagation delay than any individual link in the original chain.

The experiment then recomputes the routing tables and determines which path is selected by ns-3's global routing mechanism.

The resulting average end-to-end delay is also compared with the delay measured in Part A.

Network Topology

The original topology consists of four autonomous systems connected in a chain:

$$AS1 \to AS2 \to AS3 \to AS4.$$

A direct peering link is added between AS1 and AS4. This produces two possible paths:

$$P_1 : AS1 \to AS2 \to AS3 \to AS4$$

and

$$P_2 : AS1 \to AS4.$$

The link configuration is:

Table 1: Link configuration

Link	Data Rate	Delay
AS1–AS2	8 Mbps	3 ms
AS2–AS3	6 Mbps	4 ms
AS3–AS4	8 Mbps	3 ms
AS1–AS4	11 Mbps	1 ms

The direct AS1–AS4 link therefore has both a higher bandwidth and a lower propagation delay than every individual link in the original chain.

Path Comparison

The original chain has three hops:

$$H_{\text{chain}} = 3.$$

The direct peering path has one hop:

$$H_{\text{direct}} = 1.$$

Thus, the direct path has two fewer hops.

The direct link also has a lower routing cost than the complete three-link path. Therefore, ns-3's global routing mechanism selects the direct path.

It is important to note that this experiment uses ns-3's global IPv4 routing mechanism and does not implement the complete BGP path-selection process used on the real Internet.

The resulting forwarding path is:

$$\boxed{AS1 \rightarrow AS4}.$$

Traffic Configuration

UDP traffic is generated between AS1 and AS4.

The forward direction is:

$$AS1 \rightarrow AS4$$

and the return traffic follows:

$$AS4 \rightarrow AS1.$$

The same traffic configuration is used as in Part A so that the measured delay can be compared directly.

Console Output

The simulation was executed using the Part B configuration. The console output records the packet transmission and reception events.

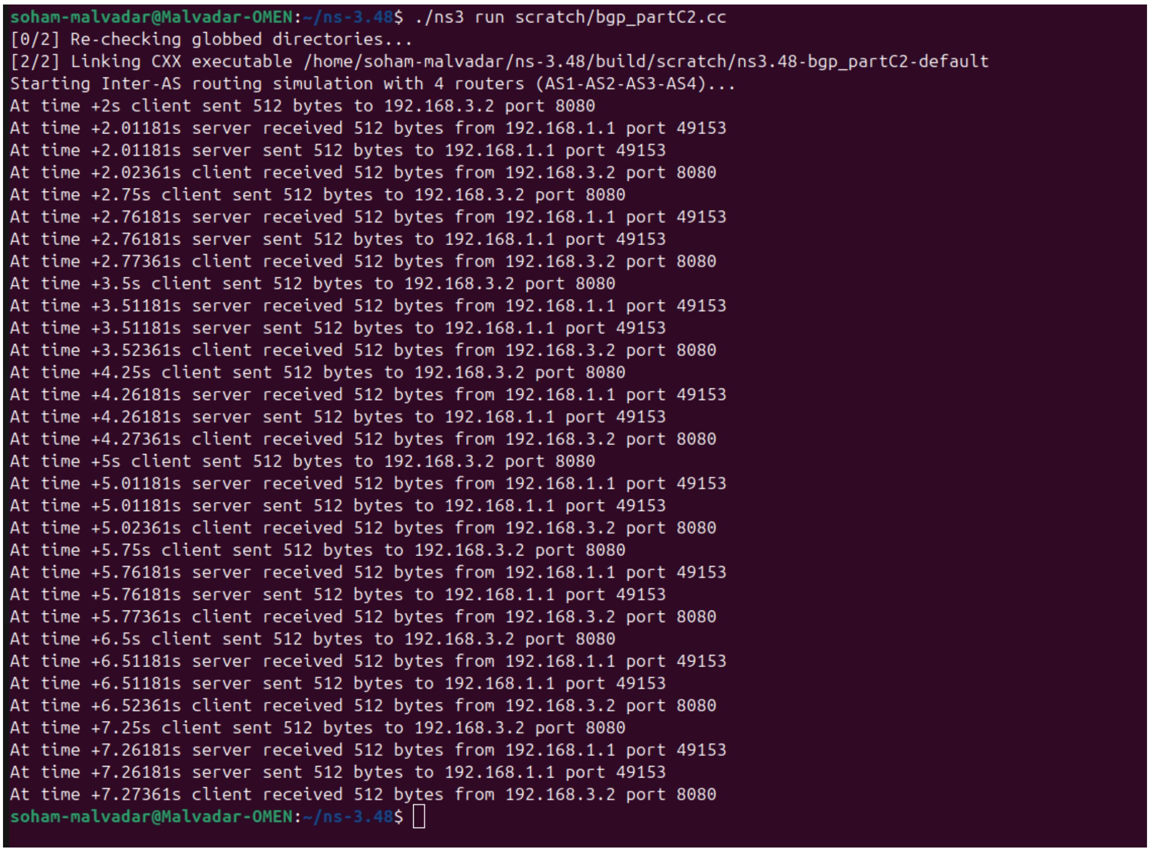

Figure 1: Console output from the Part B simulation.

The output confirms successful communication between AS1 and AS4.

NetAnim Visualization

The following screenshots show the topology and packet transmission during the Part B simulation.

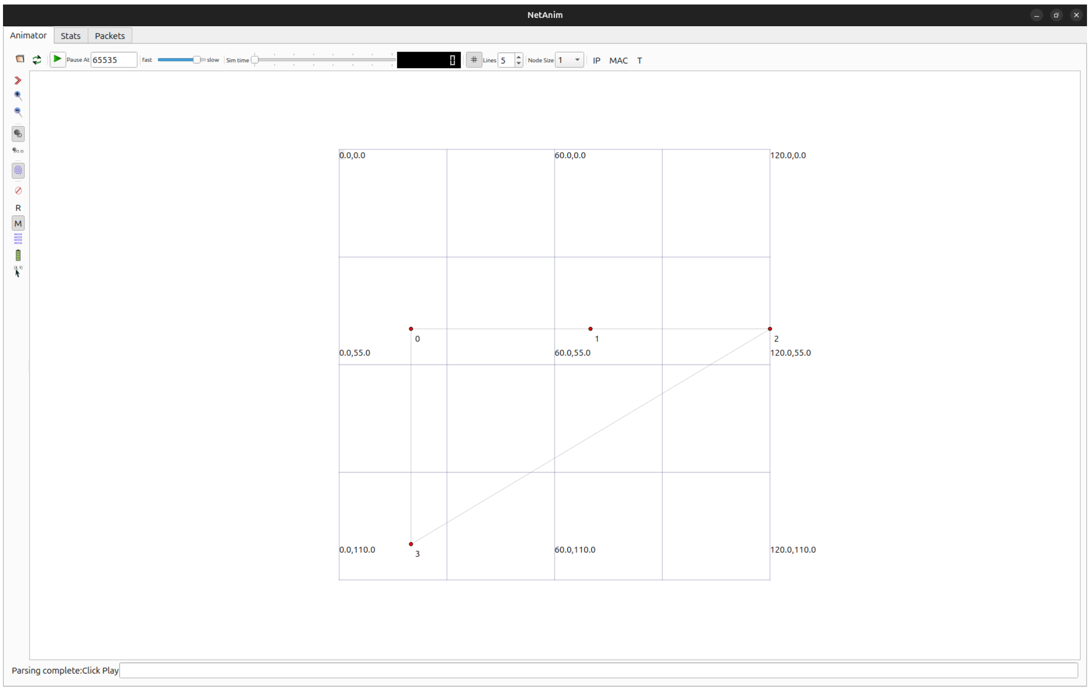

Figure 2: Part B topology showing the direct AS1–AS4 link.

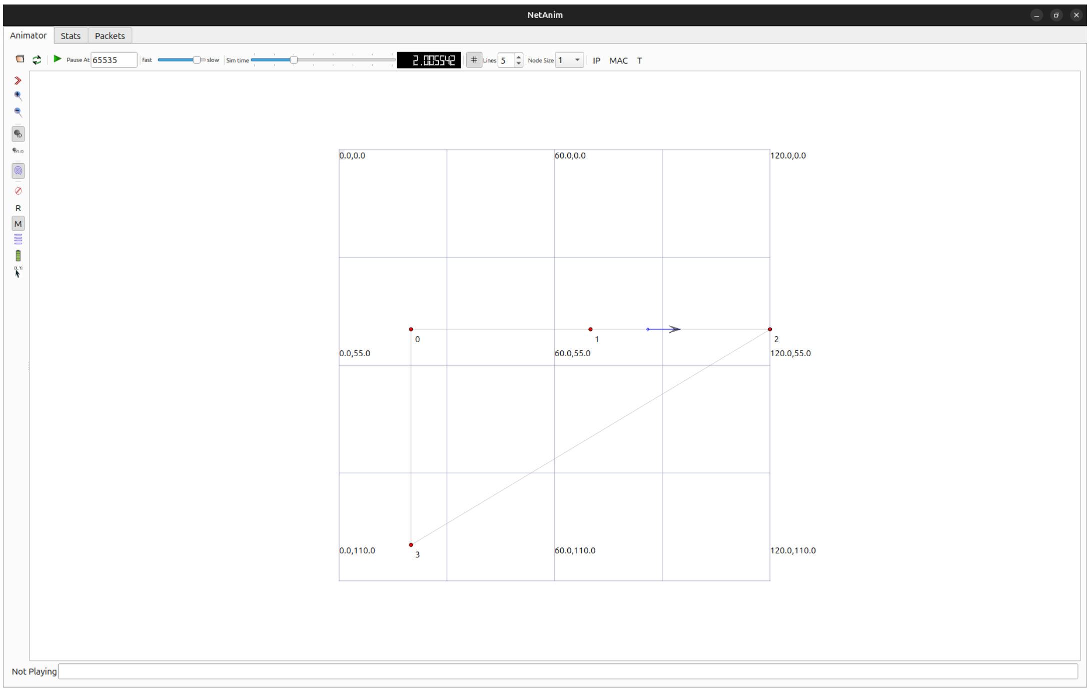

Figure 3: NetAnim frame showing packet transmission.

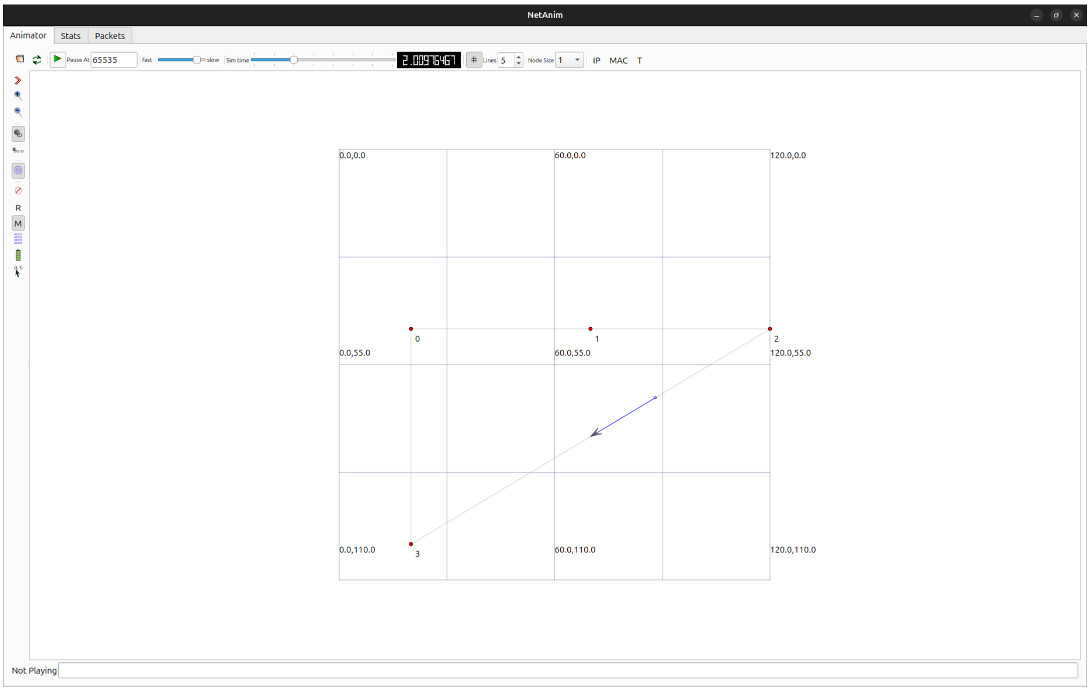

Figure 4: NetAnim frame showing packet movement between AS1 and AS4.

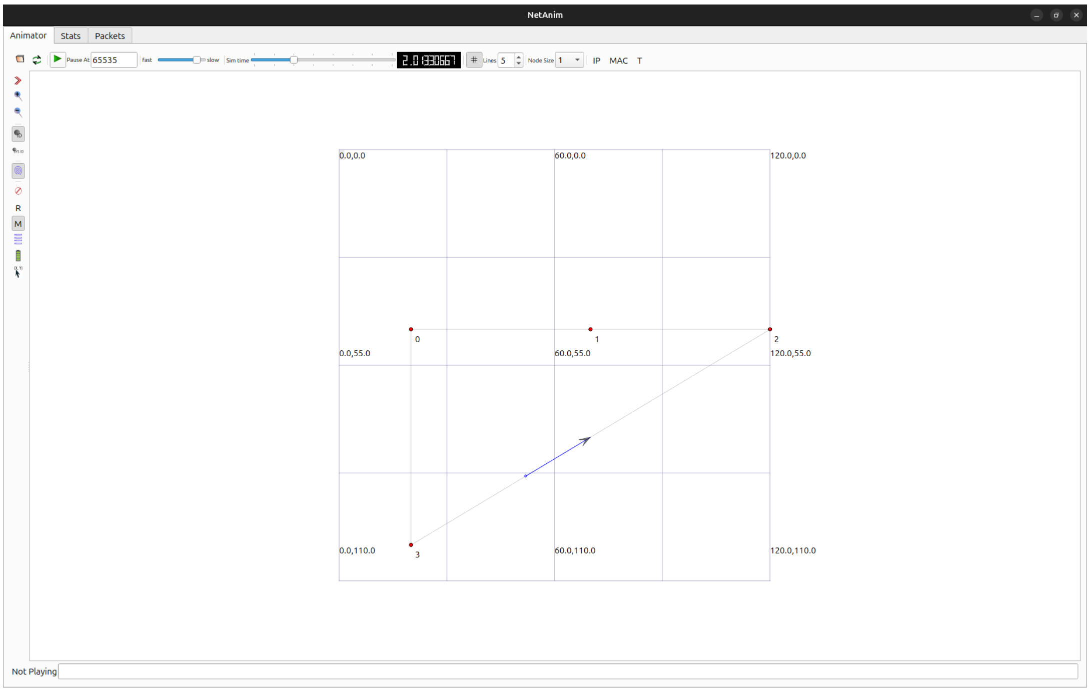

Figure 5: NetAnim frame showing continued packet transmission.

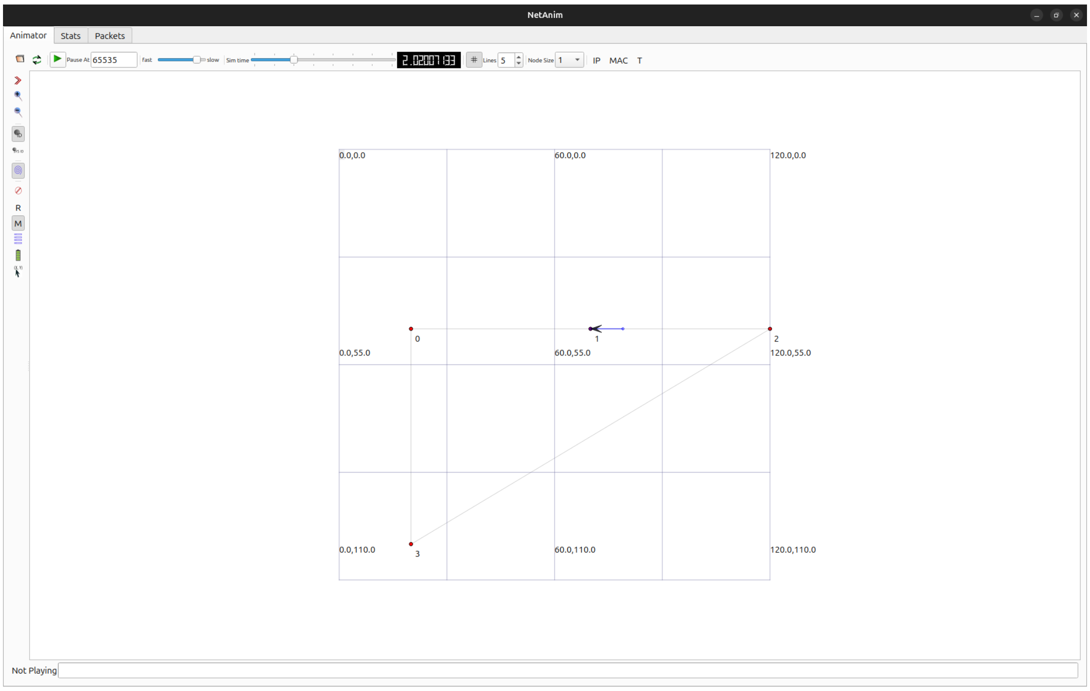

Figure 6: NetAnim frame showing the Part B packet flow.

The animation provides visual evidence of the topology and packet activity after the direct peering link is introduced.

FlowMonitor Results

The FlowMonitor XML reports the following results for the forward flow:

Table 2: Part B FlowMonitor results

Parameter	Value
Tx Packets	8
Rx Packets	8
Lost Packets	0
Times Forwarded	0

Therefore, the packet delivery rate is

$$\frac{8}{8} \times 100 = 100\%.$$

The packet loss is

$$0\%.$$

Average End-to-End Delay

The average delay is calculated from the FlowMonitor XML using

$$D_{\text{avg}} = \frac{\text{delaySum}}{\text{rxPackets}}.$$

For Part B, the XML gives

$$\text{delaySum} = 1.11535 \times 10^7 \text{ ns}$$

and

$$\text{rxPackets} = 8.$$

Therefore,

$$D_B = \frac{1.11535 \times 10^7}{8} = 1.3941875 \times 10^6 \text{ ns}.$$

Converting to milliseconds:

$$\boxed{D_B \approx 1.3942 \text{ ms}}.$$

The measured delay is very close to the expected delay of the direct link. The direct link has a propagation delay of 1 ms and a transmission delay of approximately 0.3927 ms for the 540-byte packet, giving an expected delay of approximately

$$1 + 0.3927 = 1.3927 \text{ ms}.$$

The small difference between 1.3927 ms and 1.3942 ms is due to simulation and protocol processing effects.

Delay Comparison with Part A

The average end-to-end delay measured in Part A was

$$D_A = 11.8067 \text{ ms}.$$

For Part B,

$$D_B = 1.3942 \text{ ms}.$$

The percentage reduction is calculated as

$$\text{Reduction} = \frac{D_A - D_B}{D_A} \times 100.$$

Substituting the measured values:

$$\text{Reduction} = \frac{11.8067 - 1.3942}{11.8067} \times 100.$$

Therefore,

$$\boxed{\text{Reduction} \approx 88.19\%}.$$

Thus, adding the direct AS1–AS4 peering link reduces the average end-to-end delay by approximately 88.19% compared with Part A.

Results Summary

Table 3: Part A and Part B comparison

Metric	Part A	Part B
Path	AS1–AS2–AS3–AS4	AS1–AS4
Number of Hops	3	1
Average Delay	11.8067 ms	1.3942 ms
Packet Delivery	100%	100%
Packet Loss	0%	0%
Times Forwarded	16	0

XML Results

The FlowMonitor results are stored in:

```
interas-baseline-results_B.xml
```

The XML contains the packet counts, delay statistics, forwarding information, and other flow-level measurements used in this analysis.

For the forward flow, the important values are:

```
txPackets = 8
rxPackets = 8
lostPackets = 0
timesForwarded = 0
delaySum = 1.11535e+07 ns
```

Observations

The main observations from Part B are:

1. A direct AS1–AS4 peering link is added to the original three-hop topology.
2. The direct link has higher bandwidth and lower propagation delay than every individual link in the original chain.
3. The direct path contains one hop, while the original path contains three hops.
4. The global routing mechanism selects the direct AS1–AS4 path.
5. The forward flow has `timesForwarded = 0`, confirming that packets do not pass through AS2 or AS3.
6. All eight transmitted packets are successfully received.
7. The measured average delay is 1.3942 ms.
8. Compared with the Part A delay of 11.8067 ms, the average delay is reduced by approximately 88.19%.

Conclusion

Part B demonstrates the effect of adding a direct peering link between AS1 and AS4.

The original path,

$$AS1 \rightarrow AS2 \rightarrow AS3 \rightarrow AS4,$$

requires three hops, whereas the new direct path,

$$\boxed{AS1 \rightarrow AS4},$$

requires only one hop.

After recomputing the routing tables, ns-3's global routing mechanism selects the direct path because it has the lower routing cost.

The FlowMonitor results show 8 transmitted packets, 8 received packets, zero packet loss, and zero forwarding events for the forward flow.

The measured average end-to-end delay decreases from

$$11.8067 \text{ ms}$$

in Part A to

$$1.3942 \text{ ms}$$

in Part B.

This corresponds to an approximately

$$\boxed{88.19\%}$$

reduction in average end-to-end delay.

Therefore, the direct peering link provides a significantly shorter and faster path between AS1 and AS4.